Supplementary methods: acute heat

Poisson analysis of daily emergency department visits in Los Angeles

Heat exposure

Let H_{it} be the daily maximum heat index for ZCTA i on day t. The mean $\bar{H}$ and sample standard deviation s_H were calculated across all available ZCTA-day heat observations in the Los Angeles analysis dataset, pooled across clusters and without population weighting. Heat was standardized and separated into chronic and acute components:

$$Z_{it} = \frac{H_{it} - \bar{H}}{s_H}, \qquad C_i = \frac{1}{T_i} \sum_{t \in \mathcal{T}_i} Z_{it}, \qquad A_{it} = Z_{it} - C_i. \tag{1}$$

Here $\mathcal{T}_i$ is the set of observed days with available heat measurements and $T_i = |\mathcal{T}_i|$. Chronic heat C_i is the ZCTA's mean standardized heat over the observed period. Acute heat A_{it} is the daily departure from that mean. Both use the same pooled SD; no cluster-specific SD is used.

Poisson model

Let Y_{it} denote the daily ED count (the variable `D2` in the code), N_{it} the population exposure, and $k(i)$ the assigned socioeconomic cluster. The model was

$$Y_{it} \mid \mu_{it} \sim \text{Poisson}(\mu_{it}), \tag{2}$$

$$\log \mu_{it} = \log N_{it} + \alpha + \gamma_{k(i)} + \theta C_i + \beta_{\text{acute},k(i)} A_{it} + m_{\text{month}(t)} + w_{\text{weekday}(t)} + f_{\text{doy}(t)}. \tag{3}$$

Population entered as a log offset. The model included cluster main effects, a shared chronic-heat coefficient, cluster-specific acute-heat slopes, categorical month and weekday effects, and a scaled, constrained second-order random walk for day of year. The cluster with the lowest mean vulnerability score was the reference. Each acute slope was the sum of the shared acute coefficient and its cluster interaction, with interaction zero for the reference:

$$\beta_{\text{acute},k} = \beta_{\text{acute}} + \delta_k. \tag{4}$$

Approximate Bayesian inference used INLA. Cluster-slope posterior marginals were obtained using INLA linear combinations, retaining covariance between the shared coefficient and interaction. Priors not explicitly specified in the analysis scripts used INLA defaults. The analysis conditioned on the supplied ED counts and fixed cluster assignments.

Posterior summaries and maps

Let Y denote all observed counts and $F_k(b \mid Y)$ the posterior cumulative distribution function of $\beta_{\text{acute},k}$. The probability map shows

$$p_k = \Pr(\beta_{\text{acute},k} > 0 \mid Y) = 1 - F_k(0 \mid Y). \tag{5}$$

This probability is calculated directly from the slope posterior, without exponentiating the coefficient. Values close to 1 indicate greater posterior support for a positive acute-heat association. Colors use a fixed, linear probability scale from 0 to 1 (Figure S1).

The incidence rate ratio (IRR) for a one-unit increase in acute heat is

$$\mathrm{IRR}_k = \exp(\beta_{\mathrm{acute},k}), \qquad \Pr(\mathrm{IRR}_k > 1 \mid Y) = p_k. \tag{6}$$

A one-unit acute increase corresponds to s_H heat-index units, holding chronic heat and the other model terms fixed. The IRR map shows posterior medians, with equal-tail 95% credible intervals in the legend (Figure S2). For posterior slope quantiles $q_{a,k}$,

$$\widehat{\mathrm{IRR}}_k = \exp(q_{0.5,k}), \qquad \mathrm{CrI}_{95\%,k} = [\exp(q_{0.025,k}), \exp(q_{0.975,k})]. \tag{7}$$

An IRR of 1 indicates no acute-heat association; an IRR above 1 indicates a higher modeled ED visit rate. Each ZCTA inherits its cluster's summary. The maps use 2020 ZCTA polygons clipped to the Los Angeles County boundary; areas without assigned summaries appear gray. These are area-level associations, not causal effects or individual-level risk estimates.

Calculation code

The following blocks reproduce the calculations in `summarise_heat_slopes()` in `analysis_functions.R`. Here `model` is the Poisson fit, and `cluster_terms[j]` names the INLA linear combination for cluster k.

Posterior probability of a positive acute-heat coefficient

```
term <- cluster_terms[j]
marginal <- model$marginals.lincomb.derived[[term]]

P_beta_gt_0 <- 1 - INLA::inla.pmarginal(0, marginal)
```

Posterior median IRR and 95% credible interval

```
beta_q <- INLA::inla.qmarginal(
  c(0.025, 0.5, 0.975), marginal
)

RR_median <- exp(beta_q[2])
RR_low    <- exp(beta_q[1])
RR_high   <- exp(beta_q[3])
```

The stored variable prefix `RR` refers to the acute-heat IRR.

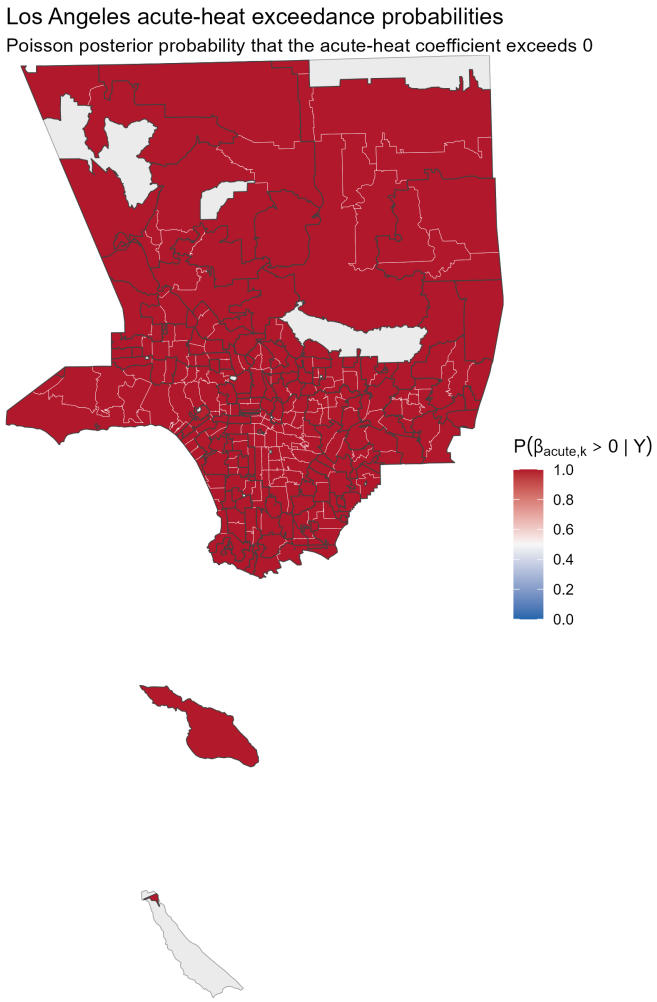


Figure S1: Poisson posterior probability that the cluster-specific acute-heat coefficient exceeds zero, $\Pr(\beta_{\text{acute},k} > 0 \mid Y)$. Each assigned ZCTA is colored by its cluster's probability.

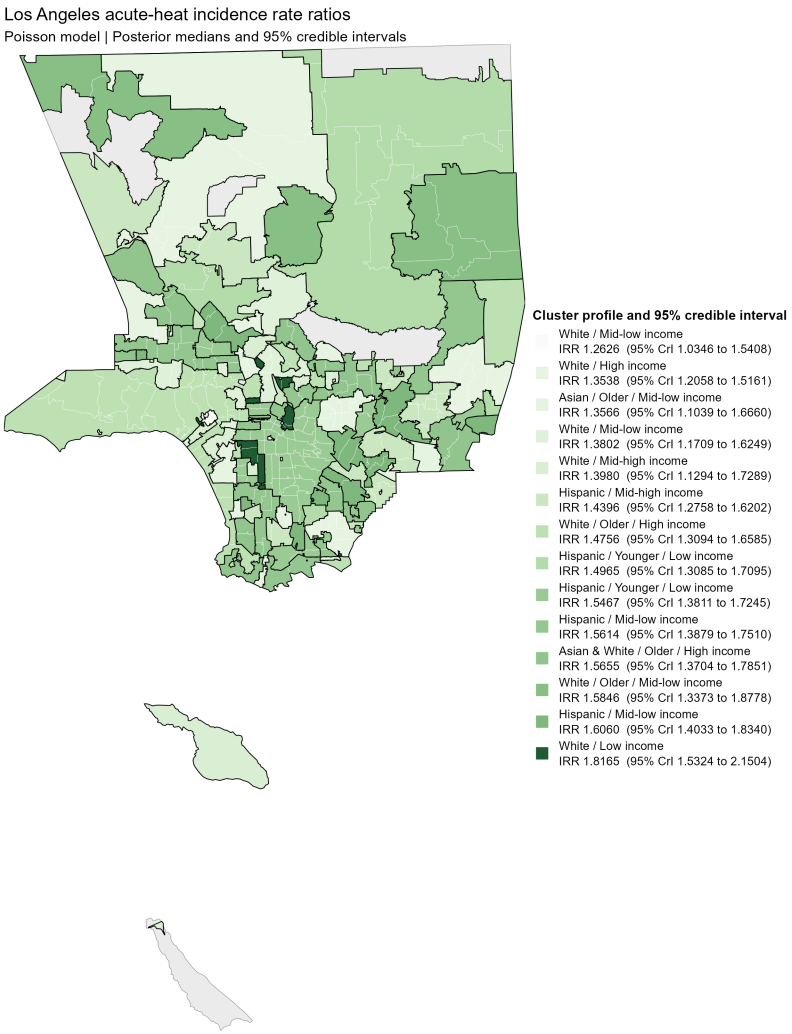


Figure S2: Poisson acute-heat incidence rate ratios per one pooled SD increase in acute heat, holding chronic heat and the other model terms fixed. Colors show posterior medians; the legend gives equal-tail 95% credible intervals.